

1. Fibonacci / Simple Recurrence

Problems where current state depends on previous 1–2 results.

Examples:

- Fibonacci Number
- Climbing Stairs
- Min Cost Climbing Stairs

2. Kadane's Algorithm (Max Subarray)

DP on subarrays tracking best-so-far.

Examples:

- Maximum Subarray
- Maximum Product Subarray
- Maximum Circular Subarray

3. Prefix DP / Running DP

Accumulate best result up to current index.

Examples:

- House Robber
- Paint House
- Delete and Earn

4. Choice DP (Take / Not Take)

Classic subset choice: take or skip.

Examples:

- Subset Sum
- Partition Equal Subset Sum
- Target Sum

5. Knapsack (0/1 & Unbounded)

Capacity-based choices of items.

Examples:

- 0/1 Knapsack
- Unbounded Knapsack
- Coin Change 2

6. DP on Subsequences / Strings

Matching subsequences or character patterns.

Examples:

- Longest Common Subsequence
- Edit Distance
- Longest Increasing Subsequence

7. Palindrome DP

DP on substrings checking palindromic structure.

Examples:

- Longest Palindromic Substring
- Count Palindromic Substrings
- Palindrome Partitioning

8. Matrix / Grid DP

DP on 2D grids using top/left transitions.

Examples:

- Unique Paths
- Min Path Sum
- Cherry Pickup

9. Interval DP

Problems where answer for $[l, r]$ depends on splitting.

Examples:

- Burst Balloons
- Matrix Chain Multiplication
- Stone Game

10. Bitmask DP

DP over subsets using masks.

Examples:

- Traveling Salesman Problem
- Count Max OR Subsets
- Assignments/Workers DP

11. Digit DP

DP across digits with constraints.

Examples:

- Count numbers with digit sum
- Count numbers without repeating digits
- Digit DP on ranges

12. Tree DP

DP from children to parent or vice versa.

Examples:

- Tree Diameter
- Max Path Sum in Tree
- Sum of Distances in Tree

13. Sliding Window + DP Hybrid

DP problems simplified using sliding window.

Examples:

- Longest Substring Without Repeating
- At Most K Distinct Characters
- Longest Substring Replacement

14. Greedy-Convertible DP

DP-like reasoning but optimized by greedy.

Examples:

- Jump Game 1
- Jump Game 2
- Activity Selection

15. Probability DP

Expected value or probability-based DP.

Examples:

- Knight Probability on Chessboard
- Dice Roll Combinations
- New 21 Game

16. Game Theory DP

Winning/losing states based on opponent move.

Examples:

- Nim Game
- Predict the Winner
- Stone Game

17. Counting / Ways DP

Counting total ways to form something.

Examples:

- Decode Ways
- Counting Paths
- Integer Partition Count

18. DP on Graphs (Topo DP)

DP on DAG using topological order.

Examples:

- Longest Path in DAG
- Shortest Path in DAG
- Course Schedule variants

19. Space Optimization DP

Reduce $dp[n][m] \rightarrow dp[m]$ or $dp[2][m]$.

Examples:

- LCS optimized
- Edit Distance optimized
- House Robber optimized